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Project: Variant Effect Dashboard

Introduction

Genetic variation is an important source of biological diversity and influences human health and disease. Single-nucleotide variations, insertions, deletions, and other types of sequence variation can all affect gene regulation, protein structure, or cellular pathways, resulting in phenotypic changes and, in rare circumstances, disease. Advances in high-throughput sequencing technologies have enabled the identification of vast numbers of genetic variants across populations. It is very important to interpret the functional and clinical implications of genetic variations. Many variations have no detectable biological effect, but some can disrupt critical genes or regulatory elements. Different information can help us identify the deleterious variants, such as genomic context, evolutionary conservation, functional predictions, population allele frequencies, and previous experimental or clinical studies. There are many tools out there that do that, but it is important to the field of genomics to continue to improve and optimize existing technology for variant prediction.

Existing Work

There are many resources available that provide high-quality annotations and partial assessments of variant impact:

- **Ensembl Variant Effect Predictor (VEP)** focuses on variant consequences at the sequence and protein level, including predicted functional effects (e.g., missense, frameshift) and scores such as SIFT, PolyPhen, and CADD.
- **gnomAD** provides large-scale population allele frequency data, allowing users to assess whether a variant is common or rare across diverse global populations.
- **Gene Ontology (GO)** databases describe the molecular functions, biological processes, and cellular components associated with genes, offering valuable biological context for interpreting gene-level effects.
- **ClinVar** aggregates clinical interpretations of variants, including classifications such as pathogenic, likely pathogenic, benign, or variants of uncertain significance (VUS), along with supporting evidence from clinicians.

These tools provide quality information but are sometimes limited. For example, VEP and other sequence-based predictors sometimes overlook biological context, molecular effect and conservation. gnomAD offers population frequency data but no clinical significance. Variant pathogenicity estimates are not directly connected to Gene Ontology annotations, which are gene-centric rather than variant-centric. Although ClinVar includes carefully chosen clinical labels, it is not comprehensive, has inconsistent interpretations for a large number of variants, and gives little assistance for ranking new or unclassified variants. Furthermore, pathogenicity predictors, like SIFT, PolyPhen, CADD, and REVEL, try to aggregate multiple signals into a single score, but they don't explicitly include gene-level functional context, such as Gene Ontology annotations. Users are frequently required to manually reconcile inconsistencies between predictors, population frequency data, and clinical annotations.

Variant interpretation remains a fragmented process, and users must switch between multiple interfaces, run custom scripts to harmonize output formats, and subjectively weigh potentially conflicting evidence. Our dashboard builds on these existing tools by treating them as complementary feature generators rather than standalone predictors. By integrating sequence-level annotations, population frequency statistics, gene-level ontology features, and clinical evidence into a unified machine-learning framework and interactive interface, we aim to provide a more coherent and user-friendly assessment of variant pathogenicity.

Objectives/Goals

1. Build an **interactive dashboard** for variant interpretation that makes variant annotation concepts approachable for those without extensive computational backgrounds
2. Combine functional predictions, allele frequency statistics, and gene-level ontology information to build an **information card for each variant**. Support exploratory analysis through searchable and visual summaries.
3. Build and train a machine learning algorithm to **more accurately estimate pathogenicity**.

Data Collection and Preprocessing

Variant- and gene-level data were collected from multiple public genomic resources, each contributing complementary information to the Variant Impact Predictor pipeline.

ClinVar

SNVs from chromosome 1 were retrieved by downloading the official ClinVar VCF file using command-line tools. The VCF was imported into R and translated to a structured tabular format

via Bioconductor functions. Variants were filtered to keep SNVs and then processed to get genomic locations, reference and alternate alleles, rsIDs, and pertinent clinical annotations.

From the VCF, we constructed a ClinVar table containing:

- Genomic position and alleles
- Variant identifiers (rsID)
- Clinical significance annotations (CLNSIG)
- Variant classification and molecular consequence fields
- Gene-level annotations (gene symbol, Entrez ID)
- Links to supporting evidence, including PubMed IDs, MedGen, OMIM, and GeneReviews entries

This table served as the primary source of labeled variants for downstream modeling and evaluation.

Ensembl Variant Effect Predictor (VEP)

ClinVar-derived genomic coordinates were utilized to generate Ensembl VEP input. Due to API size limits and timeouts when querying large variant batches programmatically, VEP annotations were collected via the VEP web interface for batch processing. This method enabled us to collect variation consequences, transcript annotations, and functional prediction scores (such as SIFT, PolyPhen, and CADD). The VEP outputs were downloaded, processed, and integrated back into the ClinVar-derived variant table via genomic position and allele matching.

Gene Ontology (QuickGO / BioMart)

Gene Ontology (GO) annotations for molecular function, biological process, and cellular component categories were obtained using getBM for VEP output gene symbols. From these annotations, we derived summary features including counts of GO terms per ontology (MF, BP,

CC) and curated indicators for biologically relevant GO categories. These features provided biological context beyond sequence-level predictions.

gnomAD

Population allele frequency data were obtained from gnomAD VCF files because they were too large to process directly through APIs. After downloading, **R (VariantAnnotation)** was used to **query the local VCF** for only your variant positions (via **GRanges**), then extract INFO fields (AF, AC/AN, population AFs, QC metrics, and optional scores like CADD/REVEL/SpliceAI where present). Derived features included global allele frequency, log-transformed frequency, and rare-variant indicators.

Additional Functional Prediction (FATHMM)

Ensemble Vep provided the prediction scores for SIFT, PolyPhen, and CADD but not Fathmm, so the VEP output was formatted and submitted to the FATHMM web interface to retrieve the score. Resulting scores were retrieved and incorporated as additional variant-level features where available.

	Location	chrom	pos	Consequence	Allele	SYMBOL	Gene	Feature	Feature_type	cDNA_position	CDS_position
1	1:69134-69134	1	69134	missense_variant	G	OR4F5	ENSG00000186092	ENST00000641515.2	Transcript	167	107
2	1:69581-69581	1	69581	missense_variant	G	OR4F5	ENSG00000186092	ENST00000641515.2	Transcript	614	554
3	1:69682-69682	1	69682	missense_variant	A	OR4F5	ENSG00000186092	ENST00000641515.2	Transcript	715	655
4	1:69731-69731	1	69731	missense_variant	C	OR4F5	ENSG00000186092	ENST00000641515.2	Transcript	764	704
5	1:69995-69995	1	69995	missense_variant	C	OR4F5	ENSG00000186092	ENST00000641515.2	Transcript	1028	968
6	1:69995-69995	1	69995	missense_variant	T	OR4F5	ENSG00000186092	ENST00000641515.2	Transcript	1028	968
7	1:925946-925946	1	925946	upstream_gene_variant	G	SAMD11	ENSG00000187634	ENST00000341065.8	Transcript	-	-
8	1:925946-925946	1	925946	missense_variant	G	SAMD11	ENSG00000187634	ENST00000342066.8	Transcript	95	5
9	1:925946-925946	1	925946	missense_variant	G	SAMD11	ENSG00000187634	ENST000003437963.5	Transcript	65	5
10	1:925946-925946	1	925946	upstream_gene_variant	G	LINC02593	ENSG00000223764	ENST00000609207.1	Transcript	-	-
11	1:925946-925946	1	925946	missense_variant	G	SAMD11	ENSG00000187634	ENST00000616016.5	Transcript	1051	542
12	1:925946-925946	1	925946	missense_variant	G	SAMD11	ENSG00000187634	ENST00000616125.5	Transcript	5	5
13	1:925946-925946	1	925946	missense_variant	G	SAMD11	ENSG00000187634	ENST00000617307.5	Transcript	5	5
14	1:925946-925946	1	925946	missense_variant	G	SAMD11	ENSG00000187634	ENST00000618181.5	Transcript	5	5
15	1:925946-925946	1	925946	missense_variant	G	SAMD11	ENSG00000187634	ENST00000618323.5	Transcript	1051	542
16	1:925946-925946	1	925946	missense_variant	G	SAMD11	ENSG00000187634	ENST00000618779.5	Transcript	5	5
17	1:925946-925946	1	925946	missense_variant	G	SAMD11	ENSG00000187634	ENST00000622503.5	Transcript	5	5
18	1:925946-925946	1	925946	downstream_gene_variant	G	-	ENSG00000272438	ENST00000756413.1	Transcript	-	-
19	1:925946-925946	1	925946	downstream_gene_variant	G	-	ENSG00000272438	ENST00000756414.1	Transcript	-	-
20	1:925946-925946	1	925946	downstream_gene_variant	G	-	ENSG00000272438	ENST00000756415.1	Transcript	-	-
21	1:925946-925946	1	925946	downstream_gene_variant	G	-	ENSG00000272438	ENST00000756416.1	Transcript	-	-
22	1:925946-925946	1	925946	downstream_gene_variant	G	-	ENSG00000272438	ENST00000756417.1	Transcript	-	-

Showing 1 to 21 of 470,000 entries, 54 total columns

Figure 1: Post-wrangling view of the unified genomic variant dataset used for downstream analysis

Table 1: Size and feature of each table and the combined dataset

Integrated Variant Feature Data			
Data Source	Unit	# Rows	Key Features
ClinVar	Variant	349665	Clinical significance, labels, review status, gene IDs
Ensembl VEP	Transcript	4427141	Consequence, SIFT, PolyPhen, CADD, protein positions
gnomAD	Variant	252463	Global + population allele frequencies, QC metrics
GO Terms	Gene	1429	BP/MF/CC gene ontology annotations
Final dataset	Merged	4726487 × 45	All features merged across ClinVar + VEP + gnomAD + GO

Programming Paradigms

The different programming paradigms that we utilized in the development of this project are explained here.

Shell scripting

ClinVar variant data were downloaded as compressed VCF files using command-line tools and subsequently imported into R for parsing and cleaning. Command-line was also used to access gnomAD release files that were not possible to retrieve through API. We downloaded gnomAD resources via both AWS S3 (`aws s3 ls/cp --no-sign-request`) and direct Google Cloud Storage downloads (`curl -O`), retrieving compressed chromosome-level “sites” VCFs along with their tabix index files (.tbi).

Object-oriented programming

In order to improve the internal logic and structure of our dashboard, we decided to create a Variant class which encompasses the genomic and annotation information we collected. Our variants have multiple fields of information associated with them, and we broadly grouped those fields into three categories: genomic, annotation, and clinical.

Table 2: Attributes in Variant S3 class

GENOMIC		ANNOTATION		CLINICAL	
Attribute Name	Information	Attribute Name	Information	Attribute Name	Information
chrom	chromosome number	gene symbol	unique gene abbreviation, via HGNC	SIFT	functional effects prediction score
pos	variant position	Ensembl_ID	unique gene identifier, via Ensembl	PolyPhen	functional effects prediction score
ref_allele	reference allele identity	strand	transcription direction (+/-)	fathmm	functional effects prediction score
alt_allele	alternate allele identity	consequence	effect of variant based on location (ex. missense, nonsense)	CADD	functional effects prediction score
				clin_sig	variant interpretation in clinical context

Encapsulating this information in a class also aids in maintainability, making it easier to add information fields if someone wants to expand the dashboard and incorporate additional databases in the future.

We chose to implement our class using the S3 system, because its straightforward approach and flexibility matched our needs well. The S3 class does not require formal class definitions, so it was well suited for constructing objects directly from single rows of our dataframe. We defined a few common functions, such as `print()` and `summary()`, and two accessor functions, `gene()` and `clinical_significance()`, to allow a user to easily access and summarize the variant information.

```
## Define S3 class: Variant
Variant <- function(df_row) {
  if (!is.data.frame(df_row)) {
    stop("Input must be a data frame")
  }
  if (nrow(df_row) != 1) {
    stop("Data frame must contain exactly one row for Variant")
  }

  obj <- list(
    genomic = list(
      chromosome = df_row$chrom,
      position = df_row$pos,
      ref = df_row$REF_ALLELE,
      alt = df_row$ALLELE
    ),
    annotation = list(
      gene = df_row$Gene,
      symbol = df_row$SYMBOL,
      transcript = df_row$STRAND,
      consequence = df_row$consequence
    ),
    clinical = list(
      phenotype = df_row$CLIN_SIG
    ),
    pathogenicity = list(
      SIFT = df_row$SIFT,
      PolyPhen = df_row$PolyPhen,
      FATHMM = df_row$fathmm,
      CADD = df_row$CADD_PHRED,
      GVPs = df_row$GVPs_pred
    )
  )

  class(obj) <- "Variant"
  return(obj)
}
```

Figure 2: Portion of code defining Variant class

Functional Programming Paradigm

Functional programming was used throughout the data processing pipeline to ensure scalable analysis. Variant annotation activities were implemented as separate functions, including generating standardized variant IDs, extracting allele frequency and functional annotation fields from nested data structures, and converting multi-valued annotations into model-ready features. These functions were then applied across several collections utilizing purrr mapping functions (e.g., `map`, `map_dbl`, `map_chr`) rather than imperative loops. For example, functional mapping was used to extract numeric, character, and logical INFO fields from compressed VCF annotation lists and to standardize multi-allelic representations into single-allele features suitable for modeling. The dashboard also uses a lot of helper functions to explore and visualize the data.

```
make_pubmed_query <- function(vr) {
  # If vr is already a gene symbol like "COL11A1"
  if (is.atomic(vr) && length(vr) >= 1) {
    gene <- as.character(vr)[1]
  } else {
    # vr is a row-like object (tibble/data.frame/list)
    gene <- ""
    if (!is.null(vr[["SYMBOL"]])) gene <- as.character(vr[["SYMBOL"]])
    if (identical(gene, "") && !is.null(vr[["hgnc_symbol"]]))
      gene <- as.character(vr[["hgnc_symbol"]])[1]
  }

  gene <- trimws(gene)
  if (is.na(gene) || !nzchar(gene)) return("")

  paste0("(", gene, "[Title/Abstract])")
}
```

```
## ----- Helper for AF donut plot -----
af_donut_plot <- function(af_value) {
  af_value <- as.numeric(af_value)
  if (is.na(af_value) || af_value <= 0 || af_value >= 1) {
    af_value <- NA_real_
  }

  df <- data.frame(
    cat = c("Alt AF", "Other"),
    value = c(af_value, 1 - af_value)
  )

  ggplot(df, aes(x = 2, y = value, fill = cat)) +
    geom_col(width = 1, color = "white") +
    coord_polar(theta = "y") +
    xlim(0.5, 2.5) +
    theme_void() +
    theme(legend.position = "bottom") +
```


Figure 3: Sample of functions used for dashboard

Machine Learning Paradigm

The machine learning (ML) component of this work aims to address this challenge by learning how to integrate diverse functional annotations into a coherent and interpretable framework. Rather than replacing existing predictors, we treat them as complementary inputs and train models to synthesize their information into robust estimates of functional severity. The result is a two-part ML system that produces both a continuous molecular severity score and interpretable categorical impact labels, which together form the foundation of the Genomic Variant Project's analytic pipeline and dashboard.

Problem Definition and Objectives

The primary scientific question motivating this work is whether machine learning models can accurately combine heterogeneous genomic annotations to predict the functional impact of genomic variants in a manner that is both quantitative and interpretable.

The original objectives of the ML component were fourfold:

- (1) to integrate diverse functional and population-level annotations into a unified dataset;
- (2) to develop and compare multiple supervised learning models for functional impact prediction;
- (3) to generate outputs that support both fine-grained ranking and categorical interpretation of variants; and
- (4) to deploy these results within a robust, user-facing data analytic product.

These objectives were achieved through the construction of a two-stage functional impact modeling framework, systematic evaluation of tree-based ML models, and deployment of trained models and predictions within an interactive dashboard environment.

Modeling Strategy: Functional Impact Prediction

The ML framework focuses on functional impact prediction, independent of clinical diagnosis. This decision reflects the biological scope of the available data and avoids overinterpretation of molecular predictions as clinical truth.

The framework is divided into two complementary tasks.

Part A: Continuous Functional Severity (GVPS Regression)

The first task predicts a continuous **Genomic Variant Pathogenicity Score (GVPS)** on a scale from 0 to 1, where lower values indicate functionally benign variants and higher values indicate increasing molecular severity. This score enables fine-grained ranking of variants within genes, pathways, or cohorts, supporting prioritization rather than binary decision-making.

Part B: Functional Pathogenicity Classification

The second task assigns each variant to one of four ordered functional impact classes:

- Benign
- Possibly Deleterious
- Probably Deleterious
- Highly Deleterious

These classes provide an interpretable abstraction of the continuous GVPS output, facilitating communication with non-technical users and downstream visualization in the dashboard.

Together, Parts A and B offer complementary perspectives: a quantitative severity spectrum and a categorical functional interpretation.

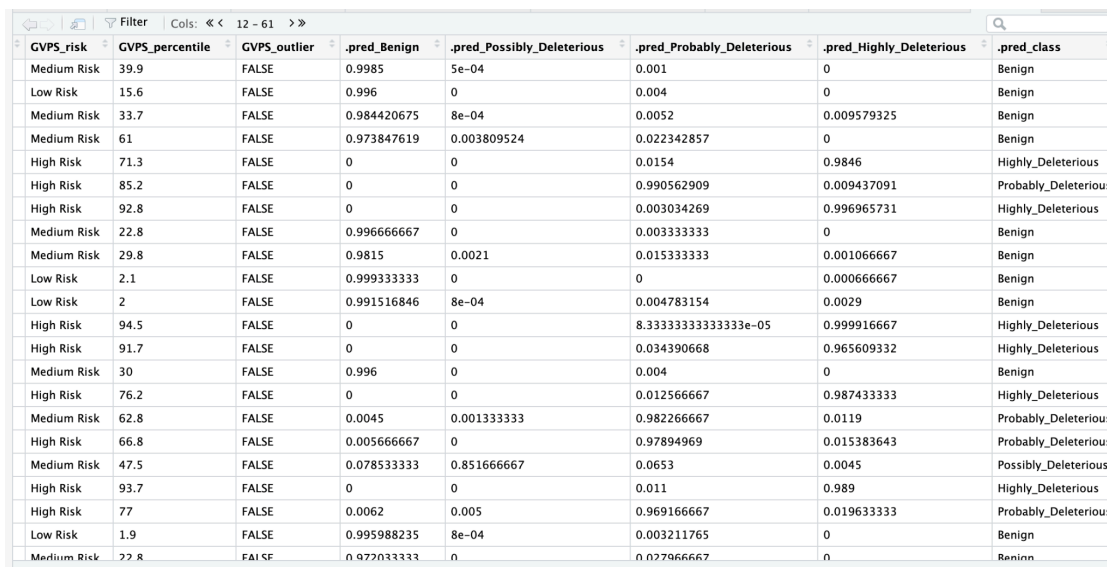
Preprocessing and Feature Engineering

Preprocessing was implemented using reproducible pipelines to ensure consistency across training, validation, testing, and deployment.

Missing numeric features were imputed using either median imputation or k-nearest neighbors, depending on feature characteristics. Categorical predictors were encoded using dummy variables, with explicit handling of unknown or missing levels to prevent silent model failures.

Continuous predictors were normalized to place heterogeneous scores on comparable scales, and zero-variance features were removed to reduce noise.

To capture nonlinear effects, optional feature transformations (such as squared CADD scores) were explored. All preprocessing steps were encapsulated in reusable recipes, enabling seamless application during both model training and real-time dashboard inference.



GVPS_risk	GVPS_percentile	GVPS_outlier	.pred_Benign	.pred_Possibly_Deleterious	.pred_Probably_Deleterious	.pred_Highly_Deleterious	.pred_class
Medium Risk	39.9	FALSE	0.9985	5e-04	0.001	0	Benign
Low Risk	15.6	FALSE	0.996	0	0.004	0	Benign
Medium Risk	33.7	FALSE	0.984420675	8e-04	0.0052	0.009579325	Benign
Medium Risk	61	FALSE	0.973847619	0.003809524	0.022342857	0	Benign
High Risk	71.3	FALSE	0	0	0.0154	0.9846	Highly_Deleterious
High Risk	85.2	FALSE	0	0	0.990562909	0.009437091	Probably_Deleterious
High Risk	92.8	FALSE	0	0	0.003034269	0.996965731	Highly_Deleterious
Medium Risk	22.8	FALSE	0.996666667	0	0.003333333	0	Benign
Medium Risk	29.8	FALSE	0.9815	0.0021	0.015333333	0.001066667	Benign
Low Risk	2.1	FALSE	0.999333333	0	0	0.000666667	Benign
Low Risk	2	FALSE	0.991516846	8e-04	0.004783154	0.0029	Benign
High Risk	94.5	FALSE	0	0	8.333333333333333e-05	0.999916667	Highly_Deleterious
High Risk	91.7	FALSE	0	0	0.034390668	0.965609332	Highly_Deleterious
Medium Risk	30	FALSE	0.996	0	0.004	0	Benign
High Risk	76.2	FALSE	0	0	0.012566667	0.987433333	Highly_Deleterious
Medium Risk	62.8	FALSE	0.0045	0.001333333	0.982266667	0.0119	Probably_Deleterious
High Risk	66.8	FALSE	0.005666667	0	0.97894969	0.015383643	Probably_Deleterious
Medium Risk	47.5	FALSE	0.078533333	0.851666667	0.0653	0.0045	Possibly_Deleterious
High Risk	93.7	FALSE	0	0	0.011	0.989	Highly_Deleterious
High Risk	77	FALSE	0.0062	0.005	0.969166667	0.019633333	Probably_Deleterious
Low Risk	1.9	FALSE	0.995988235	8e-04	0.003211765	0	Benign
Medium Risk	27.8	FALSE	0.972033333	0	0.027966667	0	Benign

Figure 4: A cleaned and standardized variant dataset after preprocessing, with all required features normalized and optimized for machine learning, and with the predicted outputs from Part A (GVPS severity score) and Part B (functional pathogenicity class) appended as new columns for downstream analysis and dashboard visualization.

Model Training and Evaluation

Data Splitting

The dataset was split into training (60%), validation (20%), and test (20%) subsets using stratification to preserve functional class distributions. This structure allowed for unbiased model selection while retaining a held-out test set for final performance assessment.

Models Evaluated

Three tree-based model families were evaluated for both regression and classification tasks:

- **Random Forest (ranger)**, providing strong baseline performance and robustness
- **XGBoost**, capable of modeling complex nonlinear interactions
- **LightGBM**, offering efficient gradient boosting with high predictive accuracy

Hyperparameters were fixed across models to facilitate fair comparison and reduce overfitting risk.

Evaluation Metrics

Regression performance (GVPS) was assessed using root mean squared error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2). Classification performance was evaluated using accuracy, balanced accuracy and macro-averaged F1 score, metrics chosen to account for class imbalance and ordinal structure.

Model selection was based on validation performance, with test-set metrics used to confirm generalization. Cross-validation metrics were logged to monitor stability and detect overfitting.

Model Performance and Outputs

Across both tasks, boosted tree models consistently outperformed baseline approaches, with strong agreement between validation and test metrics, indicating good generalization. Feature importance analyses revealed that established predictors, particularly CADD, REVEL, PolyPhen,

and SpliceAI, dominated model decisions, while conservation scores provided additional context. Population frequency features were not shown to be major contributors in the feature importance.

The trained models were serialized and saved as reusable artifacts. Prediction outputs include variant-level GVPS scores, functional class labels, and class probabilities, all of which are stored in versioned datasets and linked within the project repository to support full reproducibility.

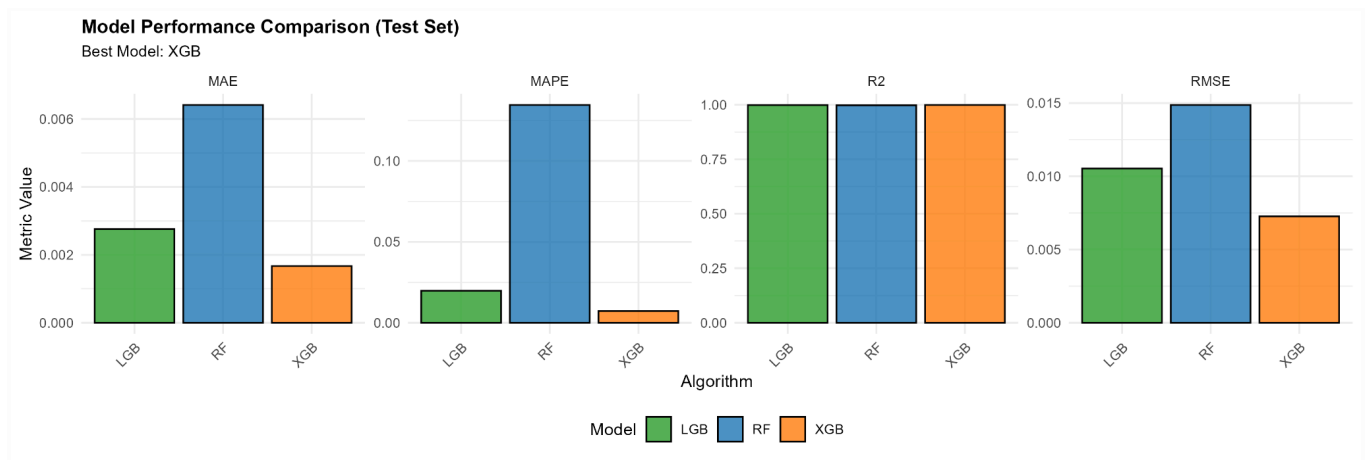


Figure 5: Model performance comparison on the GVPS regression

Table 3: Model comparison on GVPS regression

	Model	RMSE	MAE	R ²	MAPE
1	RF	0.014872717	0.006415234	0.9975515	0.134617040
2	XGB	0.007262766	0.001670732	0.9994093	0.007245536
3	LGB	0.010528895	0.002761343	0.9987597	0.019752455

Integration into the Data Analytic Product

All ML outputs are integrated into an interactive Shiny dashboard, which serves as the project's primary data analytic product. The dashboard allows users to filter variants by gene,

consequence, allele frequency, or severity thresholds; visualize GVPS distributions and functional class breakdowns; rank variants by predicted molecular impact; and inspect per-variant score contributions using normalized impact gauges.

Importantly, the dashboard is designed defensively. Optional model artifacts and explainability files are loaded dynamically, and their absence does not cause application failure. This design ensures robustness while supporting extensibility as additional models or analyses are added.

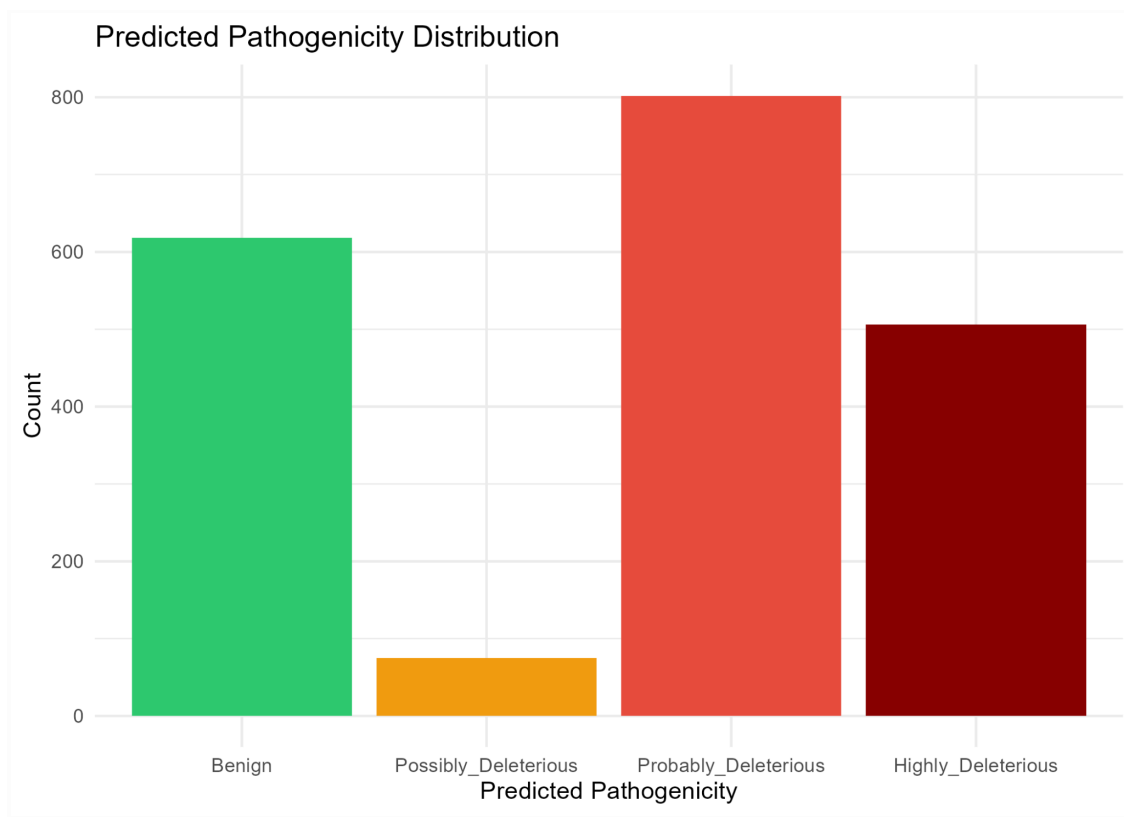


Figure 6: Distribution of variant classes in the dataset, highlighting class imbalance

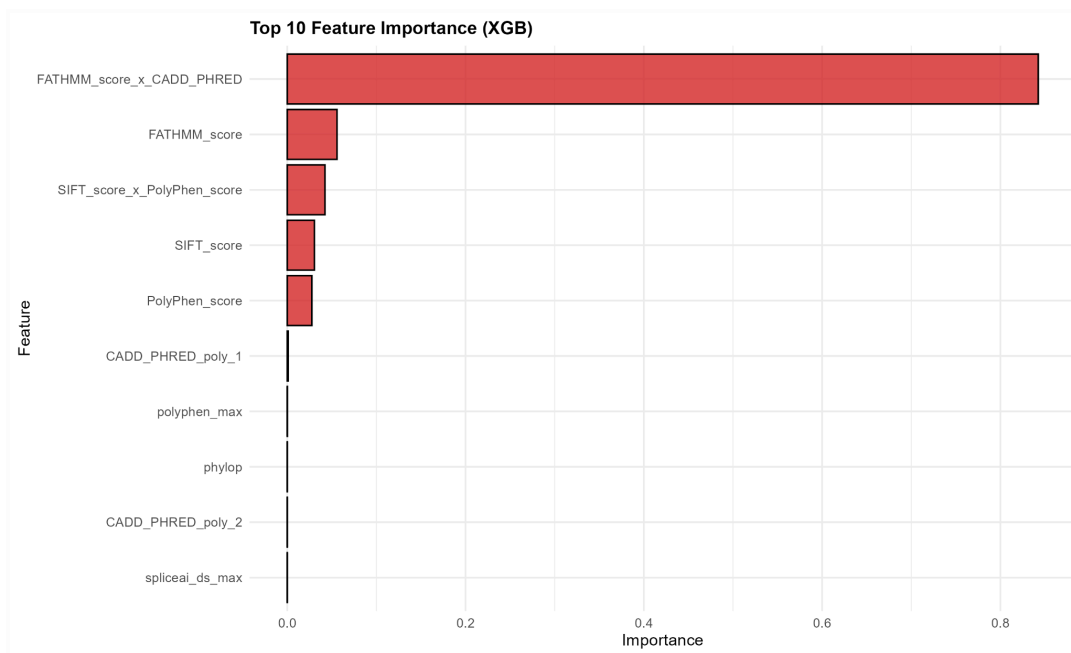


Figure 7: Feature importance plot for the GVPS model, showing the most predictive genomic features

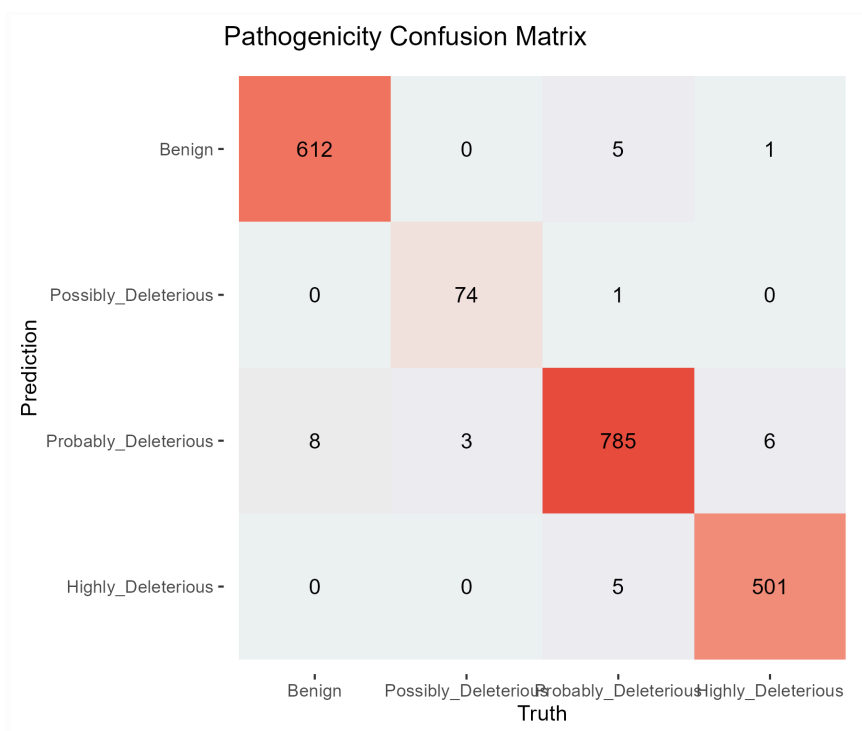


Figure 8: Confusion matrix for the pathogenicity classifier on the test set

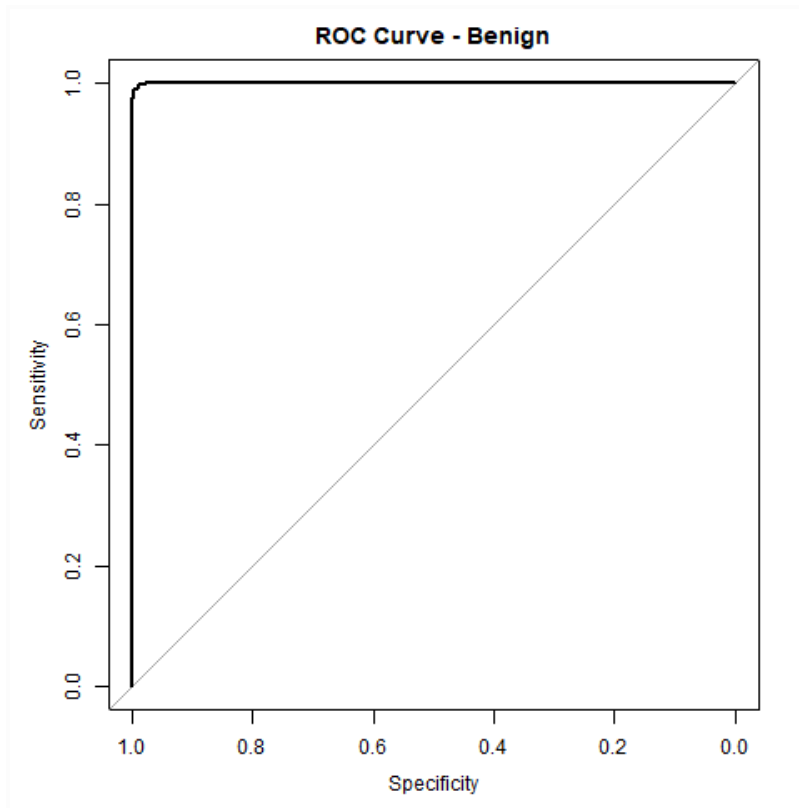


Figure 9: ROC curve for the benign variant subset, demonstrating high model discrimination

Paradigm Integration and Software Design

The ML component exemplifies the integration of multiple programming paradigms. Functional programming principles govern preprocessing, prediction, and plotting logic. Object-oriented concepts are employed through model objects, workflows, and recipes. Data-oriented programming underpins tidy data transformations, while reactive programming enables dynamic user interaction within the dashboard. This separation of concerns improves code clarity, reproducibility, and long-term maintainability.

Dashboard Overview & Functionality

Variant Information Page

The Variant information page is the core interpretation interface for our variant impact predictor.

It combines genomic context, functional annotations, population frequency statistics, gene ontology information, and literary evidence into a single, interactive presentation of a specific variant. Users enter a chromosome and genomic position to retrieve specific variant information in mini cards:

Genomic Location card: The genomic location panel visualizes the variant's position along the chromosome with the surrounding context. It also has a copyable genomic interval to explore more of the variant in genome browsers such as IGB.

Functional Risk Scores: Functional prediction scores (e.g., phyloP, REVEL, SpliceAI) are displayed on standardized scales to summarize evolutionary conservation, predicted pathogenicity, and splicing impact.

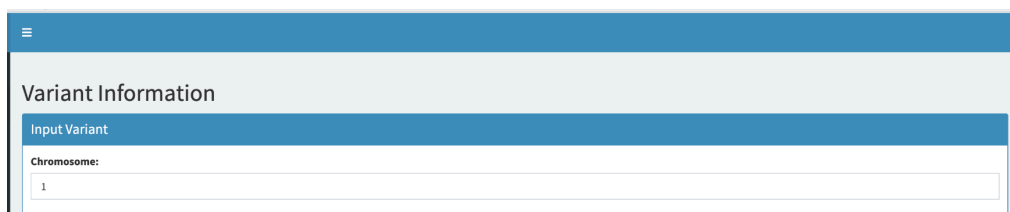
Molecular Context: This card has transcript- and protein-level annotations, including amino acid change, codon substitution, and protein position. These details help assess whether the variant affects critical or conserved regions.

Allele Frequency: Population allele frequency from gnomAD is visualized to indicate whether the variant is common or rare. This supports interpretation by contextualizing predicted impact with population prevalence.

Gene Ontology Overview: Gene Ontology annotations are summarized across biological process, molecular function, and cellular component categories.

Literature (PubMed): Relevant PubMed articles are fetched on demand and displayed with direct links for further review and help integrate experimental and clinical evidence to computational predictions.

Clinvar: The ClinVar indicator is color-coded to enable users to quickly assess the current clinical consensus while identifying cases with uncertain or conflicting interpretations.



The screenshot shows a web interface for variant information. At the top is a blue header bar with a white hamburger menu icon. Below the header is a light gray section titled "Variant Information". Under this title is a blue bar labeled "Input Variant". Below the blue bar is a form with a label "Chromosome:" and a text input field containing the number "1".

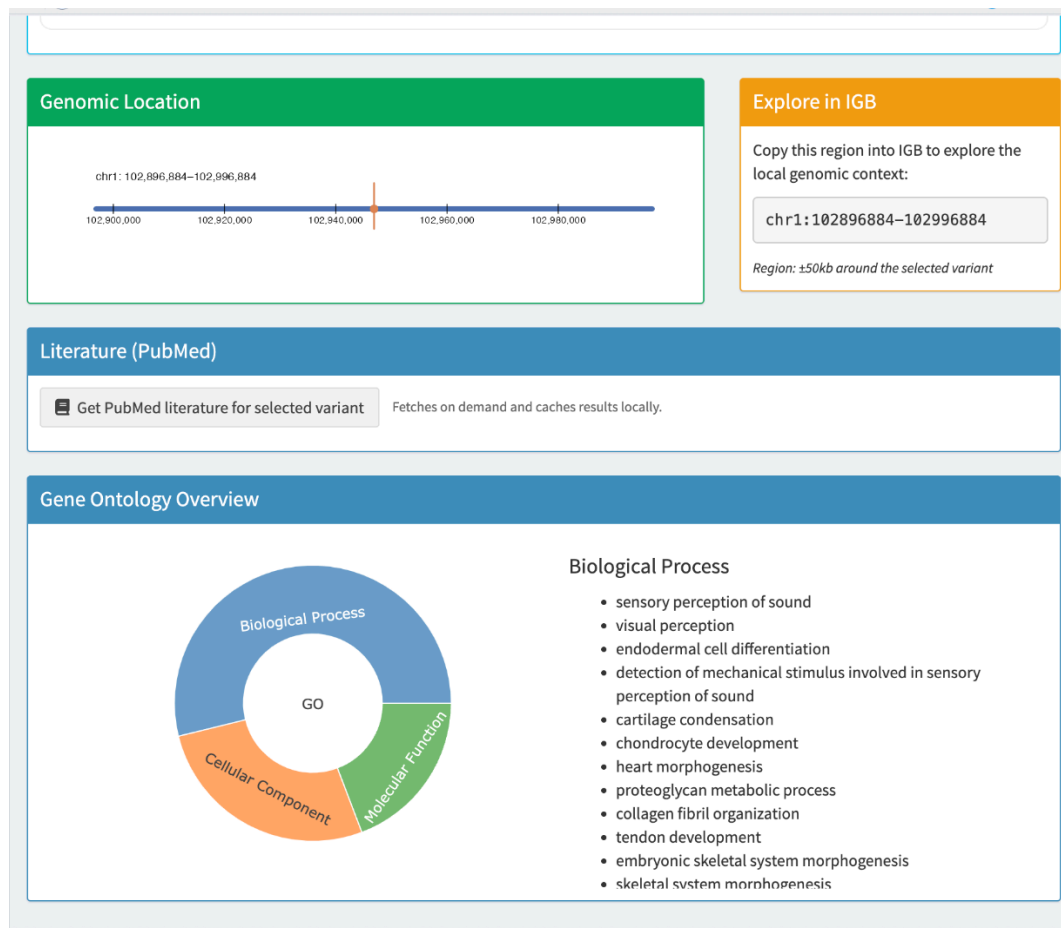
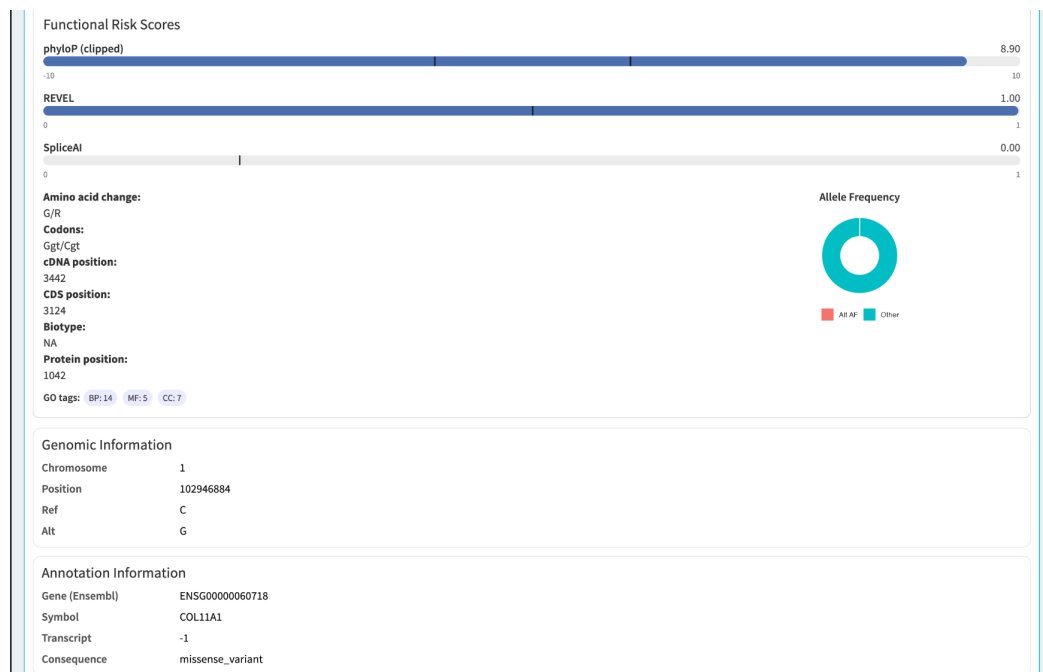


Figure 10: Image of variant info page

Variant Impact Predictor Page

The Variant Effect Prediction page summarizes computational evidence for the functional impact of a selected genetic variant. It reports pathogenicity scores from multiple widely used in silico tools, including SIFT, PolyPhen, FATHMM, CADD, and our machine learned model GVPS, each reflecting different biological assumptions such as evolutionary conservation, protein structure, and integrative deleteriousness modeling. In addition to displaying individual tool scores, the page provides a consensus interpretation that aggregates evidence across methods to classify variants as likely pathogenic, likely benign, or uncertain, along with an estimate of confidence based on inter-tool agreement. This combined view supports a more informed interpretation of variants.

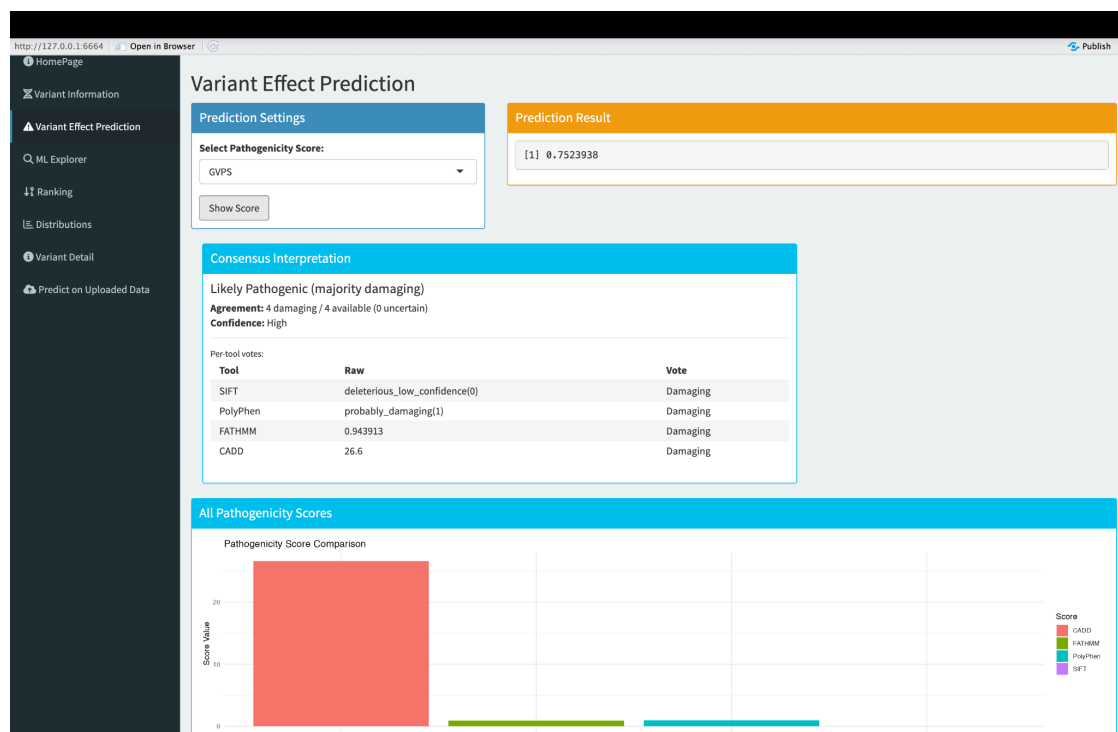


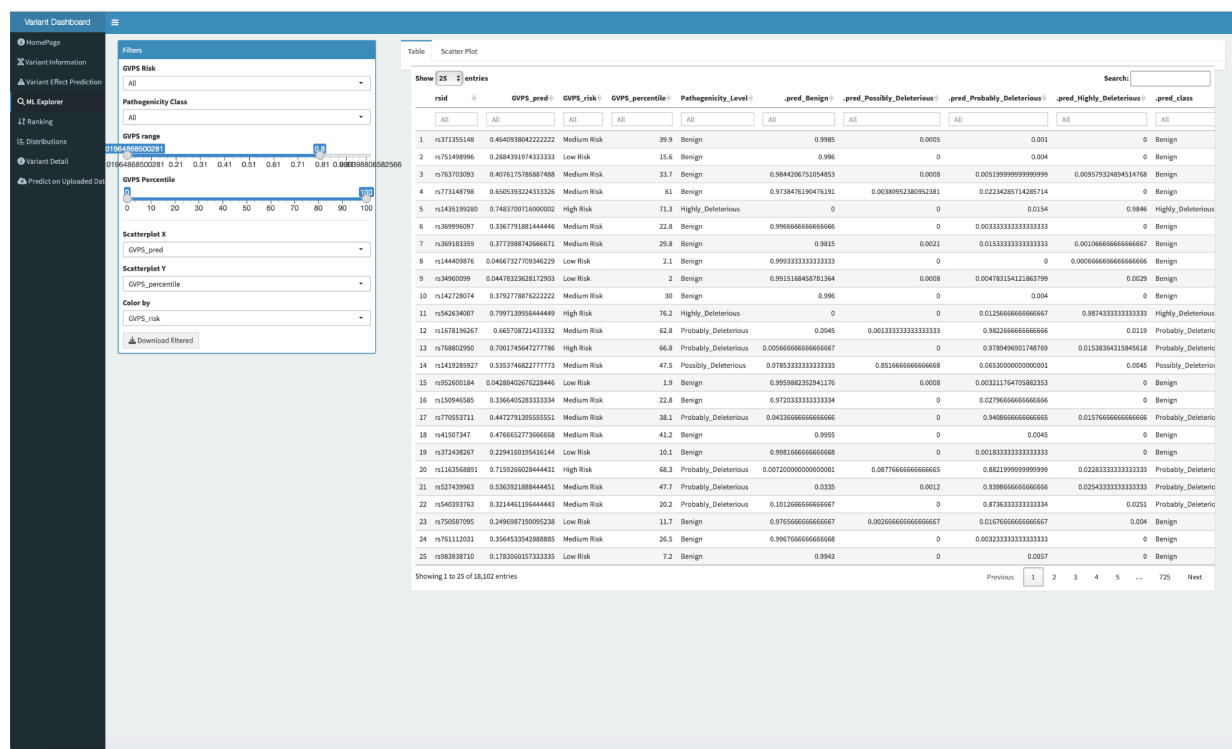
Figure 11: Tool-specific pathogenicity scores shown on their native scales

Variant Feature Explorer Pages

We have multiple pages dedicated to exploring the features of our model:

Machine Learning Explorer Page

This view is designed to support broad exploration of the full set of machine learning–predicted variants, enabling users to move fluidly from high-level patterns to specific subsets of interest. Through the sidebar controls, users can filter variants by predicted GVPS risk, functional pathogenicity class, GVPS score range, and percentile thresholds. Additional controls allow customization of scatter plot axes and color mappings, making it easy to explore relationships between features and model outputs. Results can be examined either as a sortable, searchable table or as an interactive scatter plot, and any filtered subset of variants can be downloaded for downstream analysis.



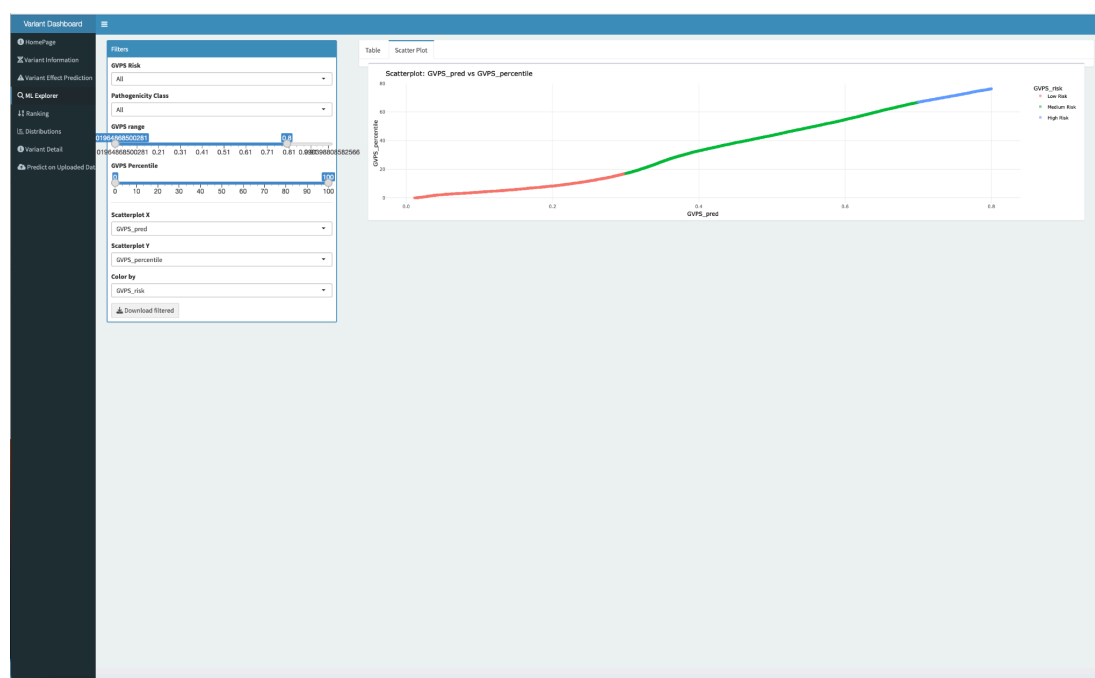


Figure 12: Screenshot of the ML Explorer page, showing interactive filtering and scatterplot visualization of machine learning predictions for genomic variants. Users can filter by risk, class, and score, and explore results in both table and plot formats

Ranking Page

This section focuses on prioritization, allowing users to quickly identify the most extreme or biologically interesting variants according to any numeric metric in the dataset. By selecting a target metric, such as GVPS, along with the desired number of variants and sorting direction, users can generate a ranked table highlighting the highest- or lowest-scoring variants. This targeted view is intended for rapid triage and focused review, supporting tasks such as candidate selection for follow-up analysis or experimental validation.

Variant Dashboard

Ranking Controls

Rank by: GVPS_pred

Top N: 25

Order: ☒ Descending ☐ Ascending

Ranking Table

Show 25 entries

Search:

	Location	chrom	pos	Consequence	Allele	SYMBOL	Gene	Feat
1	1:156873819-156873819	1	156873819	missense_variant	C	NTRK1	ENSG00000198400	ENST
2	1:78080816-78080816	1	78080816	missense_variant	C	GIPC2	ENSG00000137960	ENST
3	1:53014895-53014895	1	53014895	missense_variant	C	SCP2	ENSG00000116171	ENST
4	1:119759963-119759963	1	119759963	missense_variant	G	HMGCS2	ENSG00000134240	ENST
5	1:155293490-155293490	1	155293490	missense_variant	T	PKLR	ENSG00000143627	ENST
6	1:154988148-154988148	1	154988148	missense_variant	C	FLAD1	ENSG00000160688	ENST
7	1:151366812-151366812	1	151366812	missense_variant	G	SELENBP1	ENSG00000143416	ENST
8	1:52904668-52904668	1	52904668	missense_variant	G	ECHDC2	ENSG00000121310	ENST
9	1:63654332-63654332	1	63654332	missense_variant,splice_region_variant	T	PGM1	ENSG00000079739	ENST
10	1:16017082-16017082	1	16017082	missense_variant	T	HSPB7	ENSG00000173641	ENST
11	1:11957897-11957897	1	11957897	missense_variant	A	PLOD1	ENSG00000083444	ENST

Figure 13: Screenshot of the Ranking page, where variants are ranked by selected metrics such as predicted impact. The table highlights the top variants for focused review and prioritization.

Distribution Page

To provide context and intuition around model behavior and data structure, this panel presents a suite of distributional visualizations for key metrics. Users can examine histograms of GVPS scores and percentiles, bar plots summarizing pathogenicity classes or risk categories, and empirical cumulative distribution functions for GVPS. Additional violin and box plots enable comparison of any numeric feature across functional classes, while dropdown menus allow users to switch between gene-level or consequence-level summaries. Together, these visualizations help reveal global trends, class separations, and potential biases in the predictions.



Figure 14: Screenshot of the Distribution page, displaying histograms, barplots, and violin/boxplots summarizing the distribution of GVPS scores, pathogenicity classes, risk levels, and feature values across the dataset.

Variant Detail Page

This view is intended for deep inspection of individual variants, supporting detailed examination once a candidate of interest has been identified. Users can search for or select a specific rsID to access a concise summary of key fields, including predicted functional impact and supporting annotations. A complete table of all available data for the selected variant is also provided, allowing users to review the full annotation context without leaving the dashboard.

file can be uploaded directly through the dashboard interface, after which predictions are generated by clicking the “Run Prediction” button. Results are organized across three tabs: raw prediction outputs, pathogenicity-focused QC summaries, and GVPS-focused QC summaries, allowing users to review both individual predictions and overall data quality. All outputs can then be downloaded for downstream analysis or reporting.

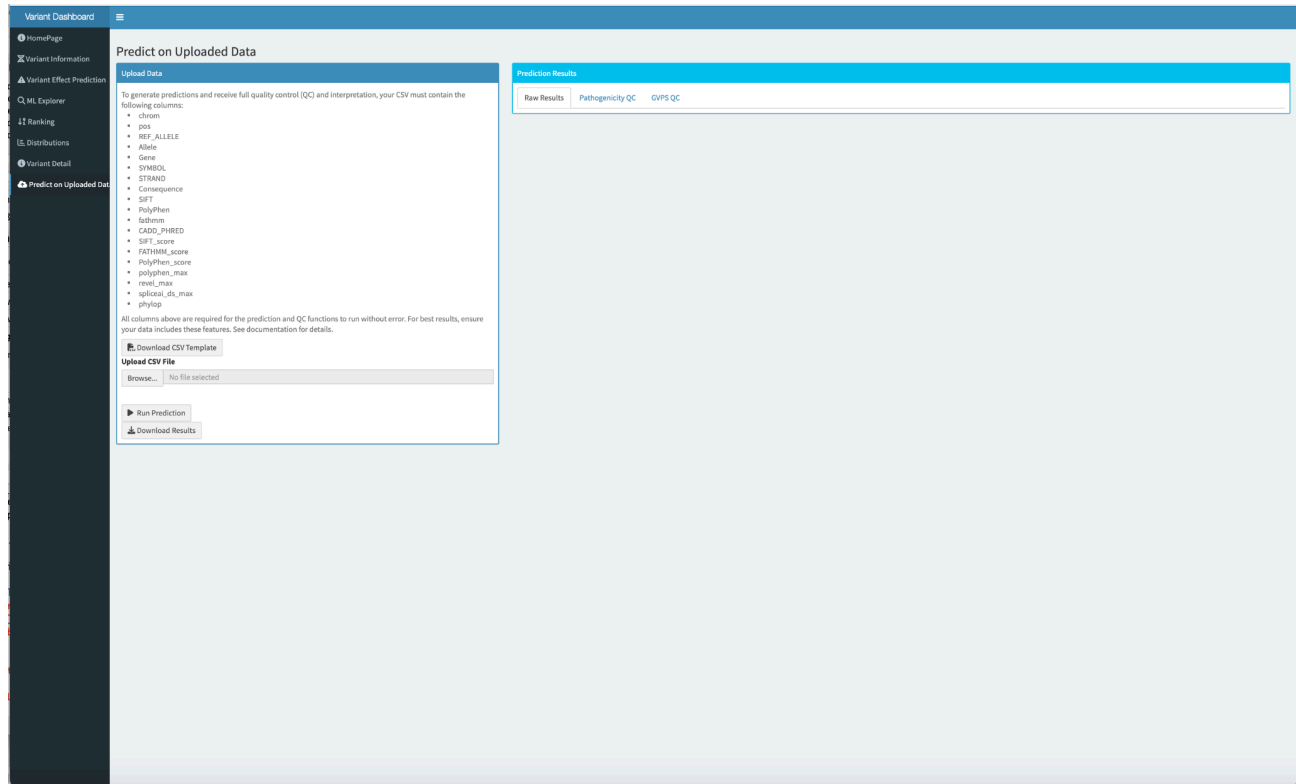


Figure 16: Screenshot of the Predict on Uploaded Data page, where users upload their own variant data, run predictions, and view results and QC summaries for their custom dataset.

Usability and Documentation

Our dashboard was designed with usability in mind, so that even untrained users are able to easily access information about their variant of interest. We have included brief instructions on the “About page” of our dashboard, and the input boxes are simple and clearly labelled. We also

present our data with prediction scores, and annotations are presented alongside their source tools and underlying values. We included external links to resources such as Ensembl, gnomAD, ClinVar, Gene Ontology, and PubMed, enabling users to verify and further investigate the evidence underlying each variant interpretation. Finally, to make our code usable as well, we have provided appropriate documentation, including a README.md file and comments throughout the code.

Challenges

We encountered several challenges in developing this dashboard.

- **Parsing and Standardization of Annotations:** Variant annotations derived from tools such as VEP often contain complex, tool-specific label formats (e.g., `benign(0.007)`), requiring careful cleaning and parsing to be able to be used as machine learning feature.
- **Missing Annotations:** Not all variants are annotated by every tool, resulting in missing values for predictors such as SIFT or REVEL. These missing annotations introduce uncertainty and limit the completeness of both visualization and modeling.
- **Data size:** The dataset contained approximately 4.7 million variants, which made preprocessing, modeling, and the interactive dashboard more challenging.
- **Data Integration:** It was a bit difficult to integrate outputs from numerous external resources and ensure consistency across identifiers, coordinate systems, and annotation formats.

Conclusion

This project has provided us with hands-on experience with end-to-end machine learning and bioinformatics tool development. Through this work, we gained experience building models using the tidymodels framework, preprocessing and normalizing heterogeneous genomic features, and handling multi-class classification with imbalanced data. We also learned how to engineer features from diverse genomic annotation tools, design interpretable machine learning outputs, and integrate predictive models into an interactive dashboard.

There are several ways this project could be expanded in the future to improve its scope and usefulness. One important next step would be extending the dashboard beyond chromosome 1 so that it can support genome-wide variant analysis and incorporate additional annotation sources. Adding live API queries from resources such as Ensembl, ClinVar, and PubMed would allow the dashboard to retrieve up-to-date information instead of relying on static datasets. Future versions could also integrate more clinical resources, such as OMIM, to provide additional context for variant interpretation. Finally, the variant classification model could be improved by adding more features from genomic tools and refining the multi-class prediction approach.

References

- Ashburner, M., Ball, C. A., Blake, J. A., Botstein, D., Butler, H., Cherry, J. M., Davis, A. P., Dolinski, K., Dwight, S. S., Eppig, J. T., Harris, M. A., Hill, D. P., Issel-Tarver, L., Kasarskis, A., Lewis, S., Matese, J. C., Richardson, J. E., Ringwald, M., Rubin, G. M., & Sherlock, G. (2000). Gene Ontology: tool for the unification of biology. *Nature*. <https://doi.org/10.1038/75556>
- Dyer, S. C., Austine-Orimoloye, O., Azov, A. G., Barba, M., Barnes, I., Barrera-Enriquez, V. P., Becker, A., Bennett, R., Beracochea, M., Berry, A., Bhai, J., Bhurji, S. K., Boddu, S., Lins, P. R. B., Brooks, L., Ramaraju, S. B., Campbell, L. I., Martinez, M. C., Charkhchi, M., . . . Yates, A. D. (2024). Ensembl 2025. *Nucleic Acids Research*, 53(D1), D948–D957. <https://doi.org/10.1093/nar/gkae1071>
- Karczewski, K. J., Francioli, L. C., Tiao, G., Cummings, B. B., Alföldi, J., Wang, Q., Collins, R. L., Laricchia, K. M., Ganna, A., Birnbaum, D. P., Gauthier, L. D., Brand, H., Solomonson, M., Watts, N. A., Rhodes, D., Singer-Berk, M., England, E. M., Seaby, E. G., Kosmicki, J. A., . . . Groop, L. (2020). The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature*, 581(7809), 434–443. <https://doi.org/10.1038/s41586-020-2308-7>
- Landrum, M. J., Lee, J. M., Benson, M., Brown, G. R., Chao, C., Chitipiralla, S., Gu, B., Hart, J., Hoffman, D., Jang, W., Karapetyan, K., Katz, K., Liu, C., Maddipatla, Z., Malheiro, A., McDaniel, K., Ovetsky, M., Riley, G., Zhou, G., . . . Maglott, D. R. (2017). ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Research*, 46(D1), D1062–D1067. <https://doi.org/10.1093/nar/gkx1153>
- Shihab HA, Gough J, Cooper DN, Stenson PD, Barker GLA, Edwards KJ, Day INM, Gaunt, TR. (2013). Predicting the Functional, Molecular and Phenotypic Consequences of Amino Acid Substitutions using Hidden Markov Models. *Hum. Mutat.*, 34:57-65